

DEGENERATE U- AND V-STATISTICS

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Summary: In this paper a partial review is given of results on degenerate U- and V-statistics. The paper focusses on asymptotic theory and includes distribution theory, invariance principles, finding the limiting distribution, the effect of estimation of unknown parameters on the limit distribution and relationships to other classes of statistics. A number of examples are discussed.

1. INTRODUCTION

Let $X_1, X_2, \dots, X_n$ be independent identically distributed (iid) random variables with distribution function F .

A U-statistic (of order 2) is given by

$$(1) \quad U_n = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} h(X_i, X_j)$$

where we assume the kernel h to be symmetric in its arguments. Clearly U_n is an unbiased estimator for

$$(2) \quad \Theta(F) \equiv E h(X_1, X_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x_1, x_2) dF(x_1) dF(x_2).$$

U-statistics were introduced by Hoeffding (1948) as a generalization of the notion of forming an average. (Hoeffding

considered the more general case of U-statistics of order m , $m \geq 1$ but for our purposes $m = 2$ will suffice. See also remark 1 in section 7.) Two well-known examples are the sample variance with $h(x_1, x_2) = \frac{1}{2}(x_1 - x_2)^2$ and the Wilcoxon statistic with $h(x_1, x_2) = I(x_1 + x_2 > 0)$.

Closely related to the above U-statistics are the V-statistics of order 2 defined by

$$(3) \quad V_n = n^{-2} \sum_{i,j=1}^n h(X_i, X_j).$$

Note that we can write (see (2))

$$(4) \quad V_n = \Theta(F_n) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x_1, x_2) dF_n(x_1) dF_n(x_2)$$

where

$$(5) \quad F_n(x) = n^{-1} \sum_{i=1}^n I(X_i \leq x)$$

is the empirical distribution function. This is an example of a differentiable statistical functional, a class of statistics introduced by von Mises (1947) (hence the V).

Assuming the existence of a second moment on h , Hoeffding (1948) showed that $n^{1/2}(U_n - \Theta(F))$ has a limiting normal distribution with mean zero and variance $4\tau^2$ with

$$(6) \quad \tau^2 \equiv \text{Var}[E(h(X_1, X_2) | X_2)] \\ = \text{Var} \left[\int_{-\infty}^{\infty} h(x, X_2) dF(x) \right]$$

if $\tau^2 > 0$. Note that the term in square brackets is the "projection" of U_n , a technique introduced and used by Hoeffding

to prove the asymptotic normality of U-statistics.

In many interesting examples (see section 2), however, $\tau^2 = 0$ because the projection is zero (or constant) and thus the asymptotic normality does not hold. In such a case the U-statistic is said to be degenerate.

Definition

A U-statistic (or V-statistic) and its kernel h are called degenerate if

$$\int_{-\infty}^{\infty} h(x, y) dF(y)$$

is a constant for almost all x .

The theory for degenerate U-statistics is much less well known than that for the usual (nondegenerate) U-statistics. In this paper a partial review is given of results for degenerate U- and V-statistics. Clearly such an overview cannot include everything and depends to an extent on the author's interests. The layout of the paper is as follows. In the next section a number of examples of degenerate U- and V-statistics are given and in sections 3 and 4 its limiting theory is given. Section 5 considers the case of U-statistics with estimated parameters and section 6 some aspects of the limiting random variable. In the final section some miscellaneous results are given.

Throughout the paper we assume $X_1, X_2, \dots, X_n$ to be a sample of iid random variables from some distribution function F . Although many of the results also hold for random vectors or more generally for random elements in some abstract space, we will confine ourselves to real random variables.

2. EXAMPLES

Many examples exist in statistics of exact or approximate (as the sample size tends to infinity) degenerate U- or V-statistics. In this section some of these will be discussed. All our examples will be of order 2; this is the most common case and although higher order degeneracy can be defined, we will restrict our attention to order 2.

2.1 Cramér-von Mises type statistics

Let F_n again denote the empirical distribution function of the sample. A well-known statistic for testing whether the sample comes from F is the Cramér-von Mises goodness-of-fit statistic given by

$$(7) \quad \omega_n^2 = n \int_{-\infty}^{\infty} (F_n(x) - F(x))^2 w(F(x)) dF(x).$$

Here w is a nonnegative weight function on $(0,1)$. Two well-known special cases are the usual Cramér-von Mises statistic with $w(u) \equiv 1$ and the Anderson-Darling statistic with $w(u) = (u(1-u))^{-1}$.

Substituting for F_n in (7), squaring and changing the order of summations and integration, give

$$(8) \quad \omega_n^2 = n^{-1} \sum_{i,j=1}^n h(X_i, X_j)$$

with

$$(9) \quad h(x, y) = \int_{-\infty}^{\infty} (I(x \leq z) - F(z))(I(y \leq z) - F(z)) w(F(z)) dF(z).$$

Thus, except for a factor n , ω_n^2 is a V-statistic of order 2. Furthermore, integrating (9) we obtain

$$\begin{aligned}
\int_{-\infty}^{\infty} h(x, y) dF(y) &= \int_{-\infty}^{\infty} (I(x \leq z) - F(z)) w(F(z)) \left[\int_{-\infty}^{\infty} (I(y \leq z) - F(z)) dF(y) \right] dF(z) \\
&= \int_{-\infty}^{\infty} (I(x \leq z) - F(z)) w(F(z)) (F(z) - F(z)) dF(z) \\
&= 0 \quad \text{for all } x.
\end{aligned}$$

Thus ω_n^2 is a degenerate V-statistic.

For some heavy tailed weight functions w , ω_n^2 does not have a limiting distribution and one has to consider instead the statistic (see Gregory (1977a) and Hall (1985))

$$n^{-1} \sum_{1 \leq i < j \leq n} h(X_i, X_j)$$

with h as in (9). This is then (essentially) a degenerate U-statistic for testing goodness-of-fit.

Two other examples of Cramér-von Mises type statistics which are degenerate are a test for symmetry and a test for independence; see Gregory (1977b) and De Wet (1980) in this regard.

2.2 Chi-square statistic

The chi-square statistic is well known as a goodness-of-fit test statistic. We now show that it is degenerate. Let $-\infty = d_0 < d_1 < \dots < d_{k-1} < d_k = \infty$ be a partition of the real line and let

$$\begin{aligned}
p_\ell &\equiv P(d_{\ell-1} < X_1 \leq d_\ell) \\
&= F(d_\ell) - F(d_{\ell-1}).
\end{aligned}$$

Estimators for p_ℓ are given by

$$\begin{aligned}\hat{p}_\ell &= F_n(d_\ell) - F_n(d_{\ell-1}) \\ &= \frac{1}{n} \sum_{i=1}^n I(d_{\ell-1} < X_i \leq d_\ell).\end{aligned}$$

with F_n as in (5).

The chi-square statistic is then given by

$$(10) \quad \frac{1}{n} \sum_{\ell=1}^k p_\ell^{-1} (\hat{np}_\ell - np_\ell)^2.$$

It follows immediately that this is (essentially) a V-statistic with

$$h(x, y) = \sum_{\ell=1}^k p_\ell^{-1} [I(d_{\ell-1} < x \leq d_\ell) - p_\ell] [I(d_{\ell-1} < y \leq d_\ell) - p_\ell].$$

Clearly $\int_{-\infty}^{\infty} h(x, y) dF(y) = 0$ for all x , and thus the chi-square statistic is degenerate. (See e.g. Denker (1985) for further results on this example.)

2.3 Functional of the empirical characteristic function

Define the empirical characteristic function (ecf) as

$$c_n(t) = \int_{-\infty}^{\infty} e^{itx} dF_n(x) = n^{-1} \sum_{j=1}^n e^{itX_j}.$$

$c_n(t)$ is an estimator of the population of

$$c(t) = \int_{-\infty}^{\infty} e^{itx} dF(x).$$

We know that F is symmetric iff $c(t)$ is real. So one can test

for symmetry of F (around 0) by using e.g.

$$\tau_n^2 = n \int_{-\infty}^{\infty} [Imc_n(t)]^2 dG(t)$$

for some G . Taking G as a distribution function symmetric around 0, we can write

$$\tau_n^2 = 2n^{-1} \sum_{i,j=1}^n [g(X_i - X_j) - g(X_i + X_j)]$$

with g the cf of G . Thus $n^{-1} \tau_n^2$ is a V-statistic and degenerate under the hypothesis that F is symmetric around 0. See e.g. Feuerverger and Mureika (1977) for further details.

2.4 Remainder in a U-statistic projection

Let h be a non-degenerate symmetric kernel and set

$$g(x) = \int_{-\infty}^{\infty} h(x,y) dF(y)$$

(the projection of h). Assume without loss of generality that $Eh(X_1, X_2) = 0$ and thus $Eg(X_1) = 0$. Define

$$\hat{h}(x,y) = h(x,y) - g(x) - g(y),$$

then $\hat{h}$ is also symmetric. Now let U_n be the U-statistic with kernel h ; then we can write

$$U_n = 2n^{-1} \sum_{i=1}^n g(X_i) + \hat{U}_n$$

with

$$\hat{U}_n = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} \hat{h}(X_i, X_j),$$

i.e. again a U-statistic. But h is clearly a degenerate kernel

and thus U_n is a degenerate U-statistic.

Thus, the remainder term in the linear approximation of a non-degenerate U-statistic is a degenerate U-statistic.

Remark

This decomposition holds for kernels of general degree m (see e.g. Serfling (1980)) and is due to Hoeffding (1961). It has recently been a strong motivational factor in studying degenerate U-statistics (see e.g. Dehling, Denker and Philipp (1984) and Nolan and Pollard (1987)).

2.5 Other examples

Degenerate U-statistics enter in the orthogonal decomposition of statistical functionals as originally studied by von Mises (1947) and also more generally in symmetric statistics (see e.g. Efron and Stein (1981)). They play a natural role in studying second order efficiency of e.g. robust statistics which are equivalent to first order (see Boos (1980)).

Silverman (1978) has also shown that certain U-statistics based on a sample of iid uniform random variables on a circle are degenerate.

3. ASYMPTOTIC DISTRIBUTION THEORY

As mentioned in section 1, if τ^2 given by (6) is positive, then $n^{1/2}(U_n - \Theta(F))$ has a limiting normal distribution with variance $4\tau^2$. If $\tau^2 = 0$, i.e. U_n is a degenerate U-statistic, then the question arises whether there is some other normalizing constant (other than $n^{1/2}$) for which U_n has a limiting distribution. In this section we show that in this case $n(U_n - \Theta(F))$ has a

limiting distribution. Since the essence of the proof is not difficult to follow, we will give an outline of it. We also briefly mention results on the rate of convergence to the limiting distribution.

For this we assume the kernel h to be square integrable, i.e.

$$(11) \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h^2(x, y) dF(x) dF(y) < \infty.$$

It then follows from the Hilbert-Schmidt theory for linear operators that there exists an eigenfunction expansion of h which converges in mean square. That is, there exists a set of complete orthonormal eigenfunctions $\{g_k\}$ and eigenvalues $\{\tau_k\}$ such that

$$(12) \quad h(x, y) = \sum_{k=0}^{\infty} \tau_k g_k(x) g_k(y)$$

where convergence is in the mean square sense, i.e.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left[h(x, y) - \sum_{k=0}^K \tau_k g_k(x) g_k(y) \right]^2 dF(x) dF(y) \rightarrow 0 \quad \text{as } K \rightarrow \infty.$$

(See e.g. Dunford and Schwartz (1963).)

Note that

$$(13) \quad \int_{-\infty}^{\infty} h(x, y) g_k(y) dF(y) = \tau_k g_k(x),$$

so by degeneracy the constant function is an eigenfunction of h and thus by orthogonality

$$(14) \quad \int_{-\infty}^{\infty} g_k(x) dF(x) = 0 \quad k = 1, 2, \dots$$

where the constant eigenfunction is g_0 . If the corresponding eigenvalue is γ_0 , then clearly $\gamma_0 = \mathfrak{e}(F)$. Without loss of generality we may take $\gamma_0 = 0$.

Theorem 3.1

Let U_n be a degenerate U-statistic with kernel h satisfying (11). Then

$$(15) \quad nU_n \xrightarrow{D} \sum_{k=1}^{\infty} \gamma_k (Z_k^2 - 1) \quad \text{as } n \rightarrow \infty,$$

where $\{Z_k\}$ are iid $N(0,1)$ random variables.

Proof

For given $\epsilon > 0$ choose K such that with

$$h_K(x, y) = \sum_{k=1}^K \gamma_k g_k(x) g_k(y),$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [h(x, y) - h_K(x, y)]^2 dF(x) dF(y) = \sum_{k=K+1}^{\infty} \gamma_k^2 < \epsilon$$

(since $\sum_{k=1}^{\infty} \gamma_k^2 < \infty$ by (11)).

Let $U_{n,K}$ be the U-statistic with kernel h_K . Then

$$\begin{aligned} U_{n,K} &\equiv \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} h_K(X_i, X_j) \\ &= (n(n-1))^{-1} \sum_{k=1}^K \gamma_k \left[\sum_{i=1}^n g_k(X_i) \right]^2 - \sum_{i=1}^n g_k^2(X_i). \end{aligned}$$

By orthogonality of $\{g_k\}$, the strong law of large numbers and the central limit theorem we have that

$$n^{-1} \sum_{i=1}^n g_k^2(X_i) \rightarrow 1 \text{ a.s., and}$$

$$n^{-1} \left[\sum_{i=1}^n g_k(X_i) \right]^2 \xrightarrow{D} Z_k^2.$$

Using these results and the Cramér-Wold device it follows that

$$(16) \quad nU_{n,K} \xrightarrow{D} \sum_{k=1}^K \gamma_k (Z_k^2 - 1).$$

Set

$$\bar{h}(x, y) = h(x, y) - h_K(x, y).$$

Since h is degenerate, it follows that

$$E[\bar{h}(X_i, X_j) | X_j] = 0 \quad \text{a.s.}$$

and thus

$$(17) \quad n^2 E(U_n - U_{n,K})^2 = 4(n-1)^{-2} E \left[\sum_{1 \leq i < j \leq n} \bar{h}(X_i, X_j) \right]^2 \\ \leq 4n(n-1)^{-1} E \bar{h}^2(X_1, X_2) \leq 8\epsilon.$$

Finally

$$(18) \quad E \left[\sum_{k=K+1}^{\infty} \gamma_k (Z_k^2 - 1) \right]^2 = 2 \sum_{k=K+1}^{\infty} \gamma_k^2 \leq 2\epsilon.$$

The theorem now follows from (16), (17) and (18).

Remarks

1. This theorem was first proved by Gregory (1977a); see also Eagleson (1979), whose proof is given here, and Hall (1979).
2. Note that the limiting distribution is characterised in terms of the eigenvalues of h ; one must thus find these. See section 6 for a further discussion of this and section 7, remark 3 for some discussion on the calculation of the

percentiles of the limiting distribution.

3. Gregory (1977a) also proved the above theorem in the case of contiguous alternatives. This changes the limiting distribution to

$$\sum_{k=1}^{\infty} \gamma_k \left[(Z_k + a_k)^2 - 1 \right]$$

with the constants $\{a_k\}$ depending on the sequence of alternatives considered.

4. A functional form of the theorem was proved by Neuhaus (1977), theorem 1.2.

Define

$$U_n(t) = \left[\frac{n}{2} \right]^{-1} \sum_{1 \leq i < j \leq [nt]} h(X_i, X_j),$$

then

$$nU_n(t) \xrightarrow{D} \sum_{k=1}^{\infty} \gamma_k \left[W_k^2(t) - t \right]$$

with $\{W_k\}$ independent Wiener processes on $(0,1)$.

5. The corresponding result to theorem 3.1 for the V-statistic is that

$$nV_n \xrightarrow{D} \sum_{k=1}^{\infty} \gamma_k Z_k^2$$

under the additional assumption that $\sum_{k=1}^{\infty} \gamma_k < \infty$.

6. For a few of the examples in section 2 the eigenvalues are known. The usual Cramér-von Mises statistic ($w \equiv 1$) has

$\gamma_k = (kH)^{-2}$, $k=1,2,\dots$ and since $\sum_{k=1}^{\infty} \gamma_k < \infty$ we can apply the result in remark 5.

For the Anderson-Darling statistic ($w(u) = (u(1-u))^{-1}$), $\gamma_k = (k(k+1))^{-1}$ so 5 can be applied again.

A weight function with very heavy tails is $w(u) = \phi(\Phi^{-1}(u))^{-2}$, with ϕ and Φ the standard normal density and distribution function respectively. For this case $\gamma_k = k^{-1}$, so that

$$\sum_{k=1}^{\infty} \gamma_k = \infty, \text{ but } \sum_{k=1}^{\infty} \gamma_k^2 < \infty.$$

Theorem 3.1 is then applicable. For the chi-square statistic there are $k-1$ nonzero eigenvalues each equal to 1 and thus the limiting distribution is χ_{k-1}^2 as is well known. (See De Wet and Venter (1973) and Gregory (1977a) for these results.)

Although quite a bit of work has been done on the rate of convergence of non-degenerate U-statistics to its limiting distributions (see e.g. Helmers and van Zwet (1982)) not too much has been done for the degenerate case. For the V-statistics, results have been obtained by Götze (1979) and Borovskikh (1984, 1985) in the degenerate case. The important thing to note here is that typically the expansion is in powers of n^{-1} rather than $n^{-1/2}$. See for example (2.13) in Götze (1979) for the case of the usual Cramér-von Mises statistic. This rapid convergence has already been noted much earlier by Marshall (1958).

4. INVARIANCE PRINCIPLES

As seen in section 3 quite a number of distribution invariance principles have been proved for degenerate U- and V-statistics. However, almost sure or in probability invariance principles have only appeared very recently, for the first time in the paper by Dehling, Denker and Philipp (1984). In this section we briefly discuss these results. It turns out that the appropriate approximating expression is a double stochastic integral with respect to the Kiefer process. We first define the latter.

Let $\{W(x,y): x \geq 0, y \geq 0\}$ be a two parameter Wiener process, i.e. a Gaussian process with $EW(x,y) = 0$ and $EW(x_1, y_1)W(x_2, y_2) = (x_1 \wedge x_2)(y_1 \wedge y_2)$ where $t_1 \wedge t_2 = \min(t_1, t_2)$.

A Kiefer process $\{K(t,y): 0 \leq t \leq 1, y \geq 0\}$ is a (separable) Gaussian process with $EK(t,y) = 0$ and covariance function $EK(t_1, y_1)K(t_2, y_2) = (y_1 \wedge y_2)(t_1 \wedge t_2 - t_1 t_2)$. It can be represented in terms of W as

$$(19) \quad \{K(t,y): 0 \leq t \leq 1, y \geq 0\} \stackrel{D}{=} \{W(t,y) - tW(1,y): 0 \leq t \leq 1, y \geq 0\}.$$

(See, e.g. Csörgö and Révész (1981).) The Kiefer process appears in a paper by Kiefer (1972) proving the first strong embedding theorem for the empirical process. In the present context it will also be used as an approximation to the empirical process. First let us mention some interesting facts about K which follow immediately from (19).

(i) For any $0 < t_0 < 1$

$$W(y) = \frac{K(t_0, y)}{(t_0(1-t_0))^{1/2}} \quad (y \geq 0)$$

is a Wiener process.

(ii) For any $y_0 > 0$

$$B(t) = \frac{K(t, y_0)}{y_0^{1/2}} \quad (0 \leq t \leq 1)$$

is a Brownian bridge.

(iii) $B_n(t) = K(t, n) - K(t, n-1)$ ($0 \leq t \leq 1$; $n = 1, 2, \dots$) is a sequence of independent Brownian bridges. Another way of saying this is that if $\{B_n(t), n = 1, 2, \dots\}$ is an independent sequence of Brownian bridges then

$$\{K(t, n): 0 \leq t \leq 1, n = 1, 2, \dots\} \equiv \left\{ \sum_{i=1}^n B_i(t): 0 \leq t \leq 1, n = 1, 2, \dots \right\}$$

defines a Kiefer process in t and n .

To show in what sense the Kiefer process approximates the empirical process we have the following.

Let $U_1, U_2, \dots, U_n$ be iid $U(0, 1)$ random variables with empirical distribution function G_n . Define the (uniform) empirical process as

$$\alpha_n(t) = n^{1/2}(G_n(t) - t), \quad 0 \leq t \leq 1.$$

Without changing the distribution of $\{U_i\}$ we can redefine it on the same probability space on which the Kiefer process below lives.

Theorem 4.1

There exists a Kiefer process $\{K(t, y): 0 \leq t \leq 1, y \geq 0\}$ such that

$$P\left(\sup_{1 \leq k \leq n} \sup_{0 \leq t \leq 1} |k^{1/2} \alpha_k(t) - K(t, k)| > (C \log n + x) \log n\right) < L \exp(-\lambda x)$$

for all x and n , where C, L, λ are positive absolute

constants. Consequently

$$\sup_{0 \leq t \leq 1} |n^{1/2} \alpha_n(t) - K(t, n)| = O(\log^2 n) \quad \text{a.s.}$$

This theorem was proved by Komlós, Major and Tusnády (1975).

See also Csörgö and Révész (1981).

Returning now to the degenerate U- and V-statistics, consider for simplicity the V-statistic. Now

$$\begin{aligned} nV_n &= n^{-1} \sum_{i,j=1}^n h(X_i, X_j) \\ &= n \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) dF_n(x) dF_n(y) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) d[\sqrt{n}(F_n(x) - F(x))] d[\sqrt{n}(F_n(y) - F(y))] \end{aligned}$$

where this last step is possible by the degeneracy of h . Applying the probability integral transformation and writing

$$(20) \quad h_0(s, t) = h(F^{-1}(s), F^{-1}(t))$$

we have

$$nV_n = \int_0^1 \int_0^1 h_0(s, t) d\alpha_n(s) d\alpha_n(t).$$

Using theorem 4.1 we can thus write approximately

$$nV_n \doteq \frac{1}{n} \int_0^1 \int_0^1 h_0(s, t) dK(s, n) dK(t, n).$$

Note that by (ii) above we can also write this as

$$nV_n \doteq \int_0^1 \int_0^1 h_0(s, t) dB(s) dB(t),$$

a result which is well known (see e.g. Filippova (1961)).

To make the above more exact we state some results below due to Dehling, Denker and Philipp (1984). To this end let

$$\|h\|_2 = \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h^2(x, y) dF(x) dF(y) \right]^{\frac{1}{2}} + \left[\int_{-\infty}^{\infty} h^2(x, x) dF(x) \right]^{\frac{1}{2}}$$

and

$$W_t = \int_0^1 \int_0^1 h_0(u, v) dK(u, t) dK(v, t), \quad t \geq 0.$$

Theorem 4.2

Assume $\|h\|_2 < \infty$. Without changing the distribution of $\{X_j\}$ we can redefine it on a new probability space on which there exists a Kiefer process $\{K(t, y); 0 \leq t \leq 1, y \geq 0\}$ such that

$$n^{-1} \max_{m \leq n} |n^2 V_m - W_m| \rightarrow 0 \text{ in probability.}$$

Furthermore, if $\|h\|_2 < \infty$ is replaced by

$$(21) \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (h(x, y) \log |h(x, y)|)^2 dF(x) dF(y) + \int_{-\infty}^{\infty} (h(x, x) \log |h(x, x)|)^2 dF(x) < \infty$$

then

$$nV_n - \frac{1}{n}W_n = o(\log \log n) \text{ a.s.}$$

For the U-statistics we have a similar result. Let

$$c = \int_{-\infty}^{\infty} h(x, x) dF(x).$$

Theorem 4.3

If $\|h\|_2 < \infty$ then there exists a Kiefer process such that

$$n^{-1} \max \left| \binom{m}{2} U_m + mc - W_m \right| \rightarrow 0 \text{ in probability.}$$

If $\|h\|_2 < \infty$ is replaced by (21), then

$$\frac{1}{2}nU_n - n^{-1}W_n = o(\log \log n) \quad \text{a.s.}$$

Remark. Law of the iterated logarithm type results for U_n and V_n are also given by Dehling, Denker and Philipp (1984). We refer the reader to their paper for details.

5. DEGENERATE U- AND V-STATISTICS WITH ESTIMATED PARAMETERS

In many instances we have statistics which would take the form of degenerate U- or V-statistics were it not that some parameter is unknown and an estimator is substituted in the statistics' form. That is, our statistics are of the form

$$(22) \quad U_n(\hat{\lambda}) = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} h(X_i, X_j; \hat{\lambda})$$

or

$$V_n(\hat{\lambda}) = n^{-2} \sum_{i,j=1}^n h(X_i, X_j; \hat{\lambda})$$

where $\hat{\lambda}$ is an estimator of some unknown parameter λ . $U_n(\lambda)(V_n(\lambda))$ is then a U-statistic (V-statistic) and degenerate if

$$\int_{-\infty}^{\infty} h(x, y; \lambda) dF(y) = \text{constant, all } x.$$

As an example consider again the problem of testing whether the sample comes from some distribution function F . However, in many cases F is not completely specified under the null hypothesis but is usually only specified up to some unknown parameters. For example, the null hypothesis might be $H_0: F(x) = F_0((x-\mu)/\sigma)$ with F_0 specified but μ and σ unknown parameters. If we then use e.g. a Cramér-von Mises type test

statistic as discussed in section 2.1, the resulting statistic will be of the form (22) with $\lambda = (\mu, \sigma)$.

In this section we discuss some results on the limiting distributions of such statistics. For the case of Cramér-von Mises statistics results were obtained by Gregory (1977a). Chi-square statistics with estimated parameters have been discussed by a number of authors. See the results of Moore and Spruill (1975) as well as their references. General results were obtained by De Wet and Randles (1987) and we will now give some of

them. If the kernel $h(x, y; \tau)$ is differentiable smooth as a function of τ , one could do a first order Taylor expansion around the true λ and in this way try to get a hold on the asymptotic distribution. However, in many cases h does not have such a smoothness property and one must resort to a different approach. To provide a general solution for a useful class of statistics without requiring the kernel to be differentiable, De Wet and Randles (1987) considered kernels of the form

$$(23) \quad h(x, y; \tau) = \int_{-\infty}^{\infty} g(x, t; \tau) g(y, t; \tau) dM(t)$$

where $M(t)$ is a finite positive measure. As an example consider again the Cramér-von Mises statistic for the hypothesis that the underlying distribution is $F_0((x-\mu)/\sigma)$ with F_0 specified but $\lambda = (\mu, \sigma)$ unknown. Let $\hat{\lambda} = (\hat{\mu}, \hat{\sigma})$ be an estimator for λ , then a Cramér-von Mises type statistic is given by

$$\begin{aligned} n \int_{-\infty}^{\infty} (F_n(\hat{\mu} + \hat{\sigma}x) - F_0(x))^2 w(F_0(x)) dF_0(x) \\ = n^{-1} \sum_{i,j=1}^n h(X_i, X_j; \hat{\lambda}) \end{aligned}$$

where

$$h(x, y; \tau) = \int_{-\infty}^{\infty} (I(x \leq \tau_1 + \tau_2 t) - F_0(t))(I(y \leq \tau_1 + \tau_2 t) - F_0(t)) w(F_0(t)) dF_0(t).$$

This is clearly of the form (23) with

$$g(x, t; \tau) = (I(x \leq \tau_1 + \tau_2 t) - F_0(t)) w(F_0(t))^{\frac{1}{2}}$$

and

$$M(t) = F_0(t).$$

Note also that h is degenerate in the sense that

$$\int_{-\infty}^{\infty} h(x, y; \lambda) dF(y) = 0, \quad \text{all } x.$$

In using the chi-square statistic (of section 2.2) for this same hypothesis we take

$$\hat{p}_\ell = F_n(\hat{\mu} + b_\ell \hat{\sigma}) - F_n(\hat{\mu} + b_{\ell-1} \hat{\sigma})$$

in (10) to obtain a statistic with kernel given by

$$h(x, y; \tau) = \sum_{\ell=1}^k p_\ell^{-1} [I(\tau_1 + \tau_2 b_{\ell-1} < x \leq \tau_1 + \tau_2 b_\ell) - p_\ell] [I(\tau_1 + \tau_2 b_{\ell-1} < y \leq \tau_1 + \tau_2 b_\ell) - p_\ell].$$

Thus, M is a discrete measure placing mass p_ℓ^{-1} on the point $t_\ell = (b_{\ell-1}, b_\ell)$, $\ell = 1, 2, \dots, k$ and

$$g(x, t_\ell; \tau) = I(\tau_1 + \tau_2 b_{\ell-1} < x \leq \tau_1 + \tau_2 b_\ell) - p_\ell.$$

Note again that $h(x, y; \tau)$ is degenerate.

In order to state our results, let

$$\mu(t; \tau) = E_\lambda g(X, t; \tau)$$

where the expectation is taken when λ is the true parameter value.

Assume that

$$\mu(t; \lambda) = 0 \quad \text{all } t.$$

This is typically the case for degenerate kernels. Denote the vector of partial derivatives of $\mu(t; \lambda)$ wrt λ by $d\mu(t; \lambda)$, assuming this to exist. For the estimator $\hat{\lambda}$ we assume a linear expansion

$$(24) \quad \hat{\lambda} = \lambda + n^{-1} \sum_{i=1}^n \alpha(X_i) + o_p(n^{-1/2})$$

where $E[\alpha(X)_r] = 0$ and $E[\alpha(X)_r \alpha(X)_s]$ is finite for all $1 \leq r \leq s \leq p$. Here u_r denotes the r -th component of a vector u and $o_p(n^{-1/2})$ is a vector whose components are each $o_p(n^{-1/2})$. In addition to these we also need a technical smoothness condition on g which will not be given (see condition 2.11 in De Wet and Randles (1987)). Suppose these assumptions hold, then we have the following result.

Theorem 5.1.

Let h_* denote the function

$$(25) \quad h_*(x, y) = \int_{-\infty}^{\infty} [g(x, t; \lambda) + d\mu(t; \lambda)' \alpha(x)] [g(y, t; \lambda) + d\mu(t; \lambda)' \alpha(y)] dM(t)$$

and suppose $Eh_*^2(X_1, X_2) < \infty$ and $Eh_*(X_1, X_1) < \infty$.

Then,

$$nV_n(\hat{\lambda}) \xrightarrow{D} \sum_{k=1}^{\infty} \delta_k Z_k^2$$

where $\{Z_k\}$ are iid $N(0, 1)$ random variables and $\{\delta_k\}$ are the eigenvalues of h_* (see (12) and (13)).

For the corresponding U-statistic we have the following result.

Theorem 5.2

Suppose the assumptions of theorem 5.1 hold and in addition that

$$E\left[\int_{-\infty}^{\infty} g^2(X, t; \lambda) dM(t)\right] < \infty.$$

Then

$$nU_n(\hat{\lambda}) \xrightarrow{D} \sum_{k=1}^{\infty} (\delta_k Z_k^2 - \nu_k)$$

where $\{\delta_k\}$ are as in theorem 5.1 and $\{\nu_k\}$ are the eigenvalues of the function $h(x, y; \lambda)$.

Remarks.

1. These results are proved in De Wet and Randles (1987).
2. Note that the effect of the estimator λ is captured in $d\mu(t; \lambda)$ (see the expression for h_* in (25)). Two possibilities arise.

If $d\mu(t; \lambda) \equiv 0$ for all t (termed Case I situations) then $nV_n(\hat{\lambda})$ and $nV_n(\lambda)$ (or $nU_n(\hat{\lambda})$ and $nU_n(\lambda)$) have the same limiting distribution and the estimator has no effect. Case II situations arise when $d\mu(t; \lambda) \neq 0$ in which case the limiting distributions are affected by the estimator λ . Examples of both types are given in De Wet and Randles (1987). There is also a third possibility where the effect of the estimator is to change the type of limiting distribution. Examples exist where the effect of the estimator is to change its limiting distribution to normal. However, such examples are rare and rather artificial.

6. OBTAINING THE EIGENVALUES

As shown in theorem 3.1, the limiting distribution of U_n is of the form $\sum_k \gamma_k (Z_k^2 - 1)$ with $\{\gamma_k\}$ the eigenvalues of h . Thus in order to find the percentiles of the limiting distribution one needs to find the $\{\gamma_k\}$. For this one has to solve the integral equation

$$\int_{-\infty}^{\infty} h(x, y) g_k(y) dF(y) = \gamma_k g_k(x).$$

This is a difficult problem to solve in a general sense. For specific choices of h one can often transform it to a differential equation which is sometimes easier to solve. As an example consider again the Cramér-von Mises statistic with kernel h given by (9). Write again

$$h(x, y) = h_0(F(x), F(y))$$

as in (20) with

$$h_0(u, v) = \int_0^1 (I(u \leq t) - t)(I(v \leq t) - t) w(t) dt.$$

For simplicity we work with h_0 and find its eigenvalues. We thus have to solve

$$(26) \quad \int_0^1 h_0(u, v) \phi_k(v) dv = \gamma_k \phi_k(u).$$

By degeneracy $\int_0^1 h_0(u, v) dv = 0$ and so by integration of (26)

$$\int_0^1 \phi_k(u) du = 0.$$

Write $\phi_k(u) = \int_0^u \phi_k(t) dt$ and substitute for h_0 in (26) to obtain

$$\gamma_k \phi_k(u) = - \int_0^u t w(t) \phi_k(t) dt + \int_u^1 (1-t) w(t) \phi_k(t) dt.$$

Differentiating this and writing $p(u) = w(u)^{-1}$ and $\lambda_k = \gamma_k^{-1}$ we see that ϕ_k satisfies the differential equation

$$(27) \quad p(u) \phi_k'(u) + \lambda_k \phi_k(u) = 0$$

with $p(u) \phi_k'(u) \rightarrow 0$ as $u \downarrow 0$ or $u \uparrow 1$.

Differentiating (27) we see that ϕ_k satisfies

$$(28) \quad dp(u) \phi_k'(u)/du + \lambda_k \phi_k(u) = 0$$

subject to the above condition.

(28) is an example of a so-called Sturm-Liouville type differential equation (see e.g. Tricomi (1970)) which for given w , one can solve for $\{\gamma_k\}$ and $\{\phi_k\}$.

For certain choices of w , the so-called classical orthogonal polynomials (Erdélyi, *et al.* (1953)) satisfy (28). We briefly indicate this; more details can be found in De Wet and Venter (1973).

The usual Cramér-von Mises statistic has $w \equiv 1$ and the Anderson-Darling statistic has $w(u) = (u(1-u))^{-1}$. In both cases the Jacobi polynomials give the solutions with eigenvalues $\gamma_k = (nk)^{-2}$ and $\gamma_k = (k(k+1))^{-1}$ respectively. The weight function $w(u) = \phi(\Phi^{-1}(u))^{-2}$, with ϕ and Φ the normal density and distribution function respectively, can be used when one is particularly concerned about the tails of the distribution. In this case the Hermite polynomials are the solutions with $\gamma_k = k^{-1}$. Finally, for the weight function $w(u) = -(1-u)^2 (\log u(1-u))^{-1}$ (see (35) in De Wet and Venter (1973) for the origin of this weight function) the solutions are given by the Laguerre

polynomials with $\gamma_k = k^{-1}$.

7. CONCLUDING REMARKS

1. In this paper we have only considered statistics of order 2. Suppose now that $U_n^{(m)}$ is a U -statistic of order m (see e.g. Denker (1985)). Then by the Hoeffding (1961) decomposition we can write

$$U_n^{(m)} = \binom{m}{2} U_n^{(2)} + R_n \quad (\text{taking } \theta(F) = 0)$$

with $E(R_n^2) = O(n^{-3})$. This means that the limiting distribution of $U_n^{(m)}$ is determined by that of $U_n^{(2)}$ and if the latter is degenerate then

$$nU_n^{(m)} \xrightarrow{D} \binom{m}{2} \sum_{k=1}^{\infty} \gamma_k (Z_k^2 - 1)$$

with $\{\gamma_k\}$ the eigenvalues of

$$h_2(x_1, x_2) = E[h(X_1, X_2, \dots, X_m) | X_1 = x_1, X_2 = x_2].$$

Similarly for V -statistics. Thus as far as limiting distribution theory is concerned there is no loss of generality in considering only $m = 2$.

2. Extensions to the two sample case and dependent observations were given by Eagleson (1979). We mention his results for the latter case.

Consider again

$$U_n = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} h(X_i, X_j)$$

where we now assume $\{X_i\}$ to be a strictly stationary sequence, uniformly strongly mixing with mixing coefficient

$\phi(n)$ (see e.g. Ibragimov and Linnik (1971)). The common marginal distribution function of the X 's is denoted by F . In order to define degeneracy recall that we have previously seen that if h is degenerate one of its eigenfunctions is constant (see section 3). In the present case we take this as definition of degeneracy since the definition does then not depend on the joint distribution of the X 's. Suppose again that h is square integrable with eigenvalues $\{\gamma_k\}$ which we assume to be such that

$$\sum_{k=1}^{\infty} |\gamma_k| < \infty.$$

Finally assume that $\phi(n)$ is such that

$$\sum_{n=1}^{\infty} \phi(n)^{1/2} < \infty.$$

Then

$$nU_n \xrightarrow{D} \sum_{k=1}^{\infty} \gamma_k (\hat{Z}_k^2 - 1) \quad \text{as } n \rightarrow \infty,$$

where $\{\hat{Z}_k\}$ is a sequence of correlated, jointly normal random variables.

3. To find the percentiles of the limiting random variable

$$\sum_{k=1}^{\infty} \gamma_k (\hat{Z}_k^2 - 1)$$

once the eigenvalues are known, different methods can be used.

It is easy to write down its characteristic function and fairly easy to invert this numerically; see Imhof (1961) and also Martynov (1975). A very useful approximation for large x of

$$G(x) = P\left[\sum_{k=1}^{\infty} \gamma_k (Z_k^2 - 1) \leq x\right]$$

is given by Gregory (1980), corollary 2.2. Another possibility is to approximate $G(x)$ by a distribution in the Pearson system, since the moments of

$$\sum_{k=1}^{\infty} \gamma_k (Z_k^2 - 1)$$

can easily be written down. See e.g. Solomon and Stephens (1978).

4. There are close connections between, on the one hand, U_n and V_n and, on the other hand, the study of quadratic forms in independent random variables. In particular, one has the same limiting random variable in both cases (see e.g. Varberg (1966) and De Wet and Venter (1973)).

Recent work on symmetric statistics also has close ties with degenerate U- and V-statistics; see e.g. Rubin and Vitale (1980), Mori and Székely (1982), Dynkin and Mandelbaum (1983), Mandelbaum and Taqqu (1984). Using the expansion (12) in a pointwise fashion we can write intuitively

$$U_n = \sum_{k=1}^{\infty} \gamma_k \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} g_k(X_i) g_k(X_j)$$

and the inner sums form what is called an elementary symmetric polynomial of order 2. The limit distribution of such polynomials is in terms of $H_2(Z)$, the second Hermite polynomial evaluated at a $N(0,1)$ random variable (Mori and Székely (1982)).

$$nU_n \xrightarrow{D} \sum_{k=1}^{\infty} \gamma_k H_2(Z_k) = \sum_{k=1}^{\infty} \gamma_k (Z_k^2 - 1)$$

with H_2 the second Hermite polynomial and $\{Z_k\}$ again iid $N(0,1)$.

5. The limiting random variable can also be written as a Wiener integral of order 2; see e.g. Hida (1980), theorem 4.8 or Dynkin and Mandelbaum (1983).

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